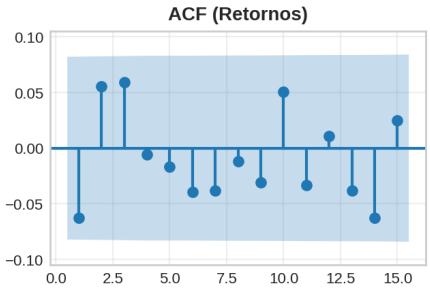
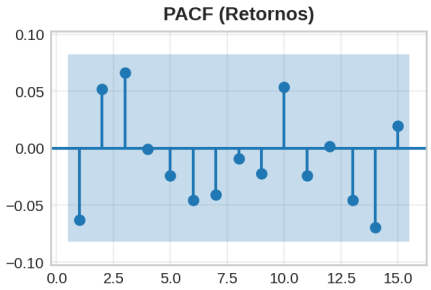
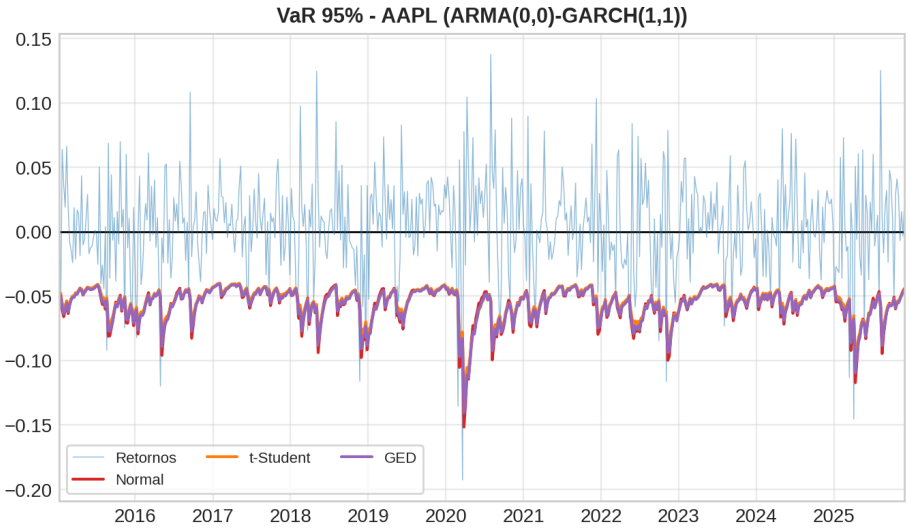
AAPL

Apple Inc.: Líder tecnológico en hardware y servicios.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con GED
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 33/28
- **Tasa de Fallo:** 5.80% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	3124	0.175	0.663	0.651	nan
t-Student	3096	0.122	0.756	0.620	5.43
GED	3095	0.142	0.718	0.653	1.29

Serie: AAPL
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.13
Volatilidad	0.28
Sharpe	0.67
VaR 95	-5.53%
VaR 99	-11.42%

Fundam.	
Price	$258.27
PE	34.4
Target	$300.0
Upgrade	15.0%

Nota: Sensible Tech.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.4368
- ARCH-LM (H.Cond): p=0.0000

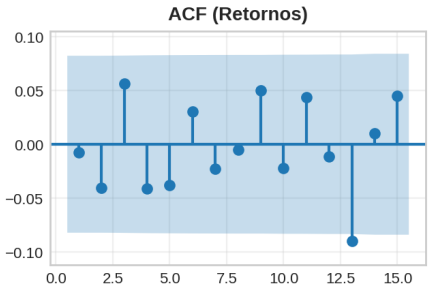
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **GED**
- Suma $\alpha + \beta$: 0.859 (<1)
- AIC: 3095
- Ljung-Box (Resid. Std.): p=0.649
- ARCH-LM (Resid. Std.): p=0.112

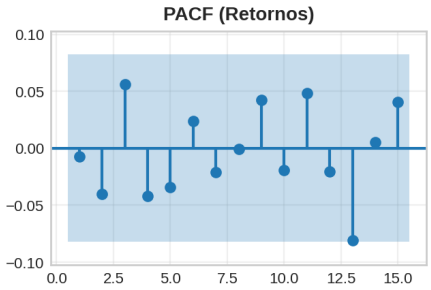
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.393
- Christoffersen (Cond.): p=0.693

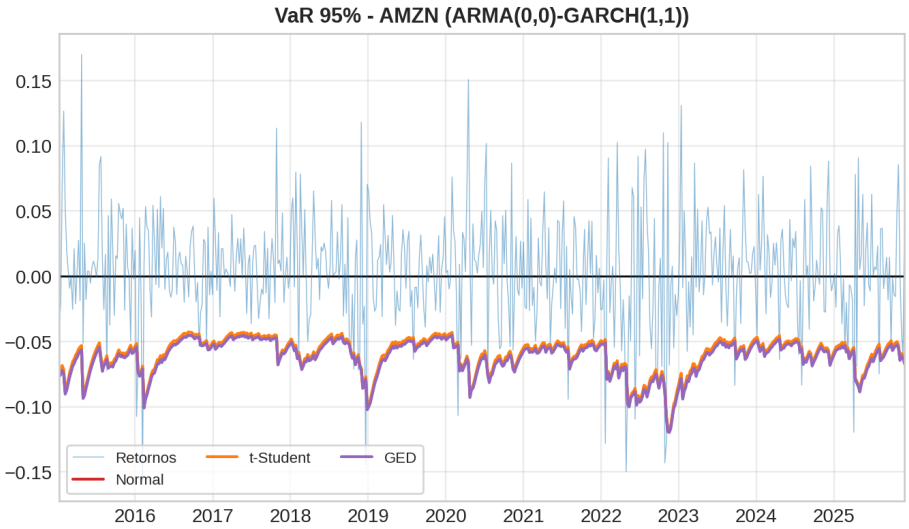
AMZN

Amazon: E-commerce y Cloud (AWS).

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 31/28
- **Tasa de Fallo:** 5.45% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	3217	0.082	0.852	0.594	nan
t-Student	3190	0.095	0.839	0.591	5.72
GED	3190	0.085	0.849	0.548	1.31

Serie: AMZN
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.07
Volatilidad	0.30
Sharpe	0.69
VaR 95	-6.29%
VaR 99	-12.23%

Fundam.	
Price	$244.68
PE	34.4
Target	$300.0
Upgrade	20.4%

Nota: Consumo Cíclico.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.7100
- ARCH-LM (H.Cond): p=0.0173

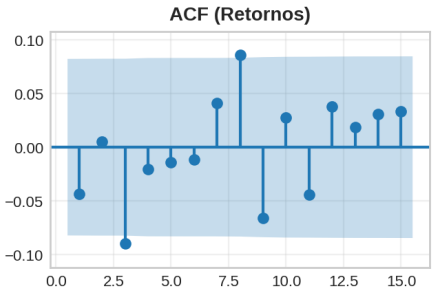
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.934 (<1)
- AIC: 3190
- Ljung-Box (Resid. Std.): p=0.958
- ARCH-LM (Resid. Std.): p=0.823

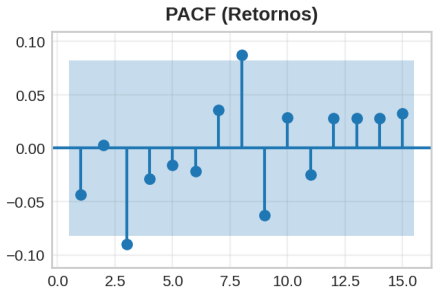
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.629
- Christoffersen (Cond.): p=0.863

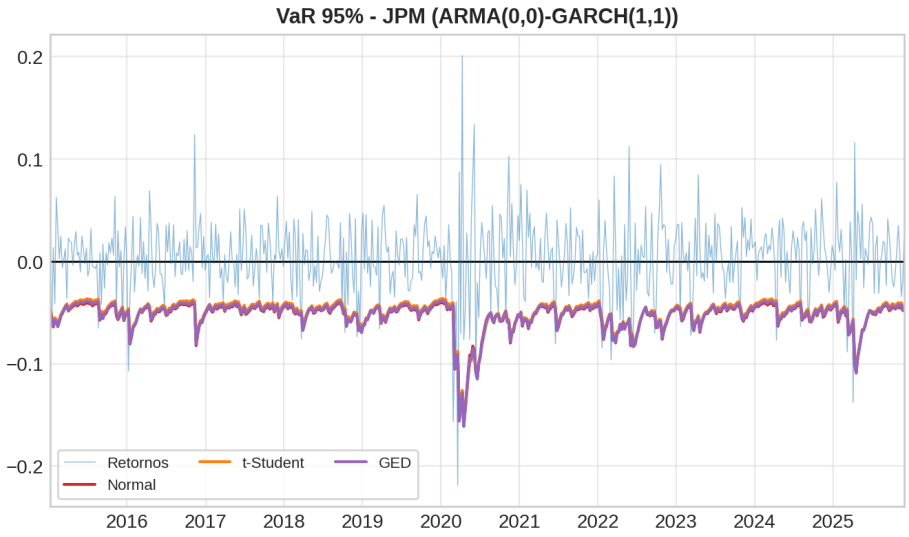
JPM

JPMorgan: Banca Sistémica Global.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 36/28
- **Tasa de Fallo:** 6.33% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	3024	0.140	0.693	0.440	nan
t-Student	2988	0.130	0.747	0.499	5.89
GED	2997	0.138	0.709	0.514	1.35

Serie: JPM
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.15
Volatilidad	0.27
Sharpe	0.51
VaR 95	-5.71%
VaR 99	-9.08%

Fundam.	
Price	$300.31
PE	15.0
Target	$345.0
Upgrade	13.9%

Nota: Ciclo Financiero.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.1549
- ARCH-LM (H.Cond): p=0.0000

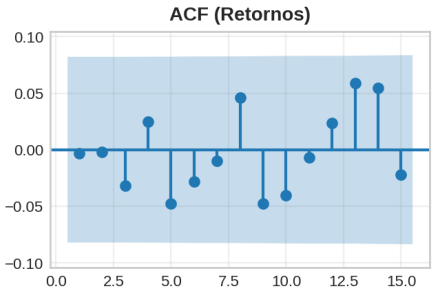
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.877 (<1)
- AIC: 2988
- Ljung-Box (Resid. Std.): p=0.858
- ARCH-LM (Resid. Std.): p=0.280

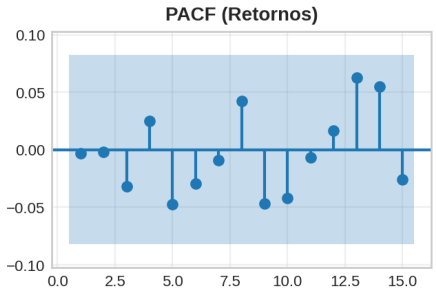
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.162
- Christoffersen (Cond.): p=0.326

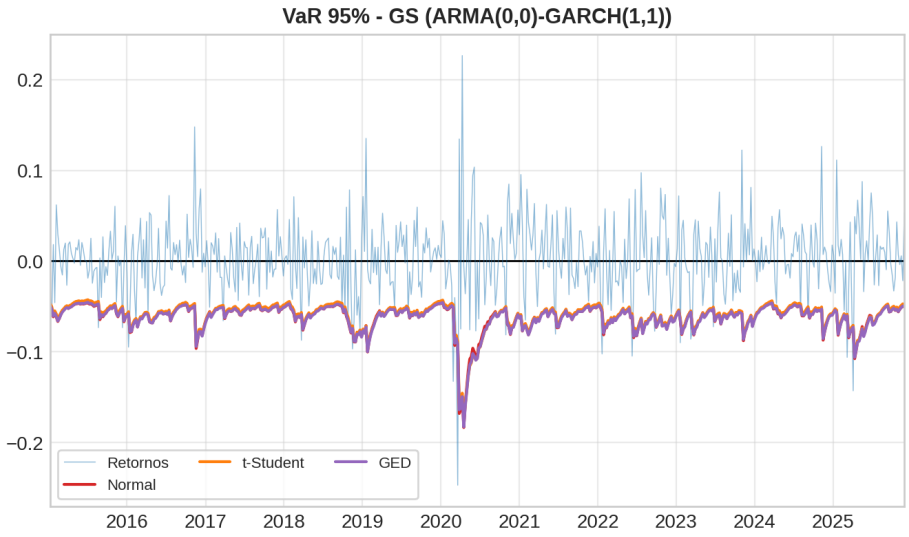
GS

Goldman: Banca de Inversión.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 31/28
- **Tasa de Fallo:** 5.45% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	3172	0.130	0.728	0.362	nan
t-Student	3142	0.118	0.781	0.335	5.92
GED	3149	0.120	0.759	0.329	1.38

Serie: GS
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.34
Volatilidad	0.30
Sharpe	0.38
VaR 95	-6.30%
VaR 99	-9.85%

Fundam.	
Price	$929.72
PE	18.2
Target	$970.0
Upgrade	4.2%

Nota: Alto Beta.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.7932
- ARCH-LM (H.Cond): p=0.0000

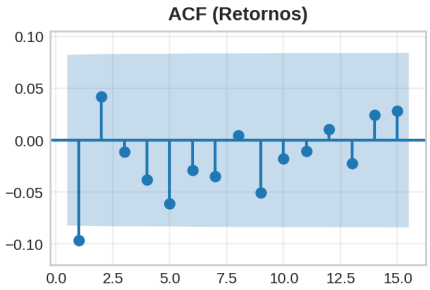
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.899 (<1)
- AIC: 3142
- Ljung-Box (Resid. Std.): p=0.890
- ARCH-LM (Resid. Std.): p=0.580

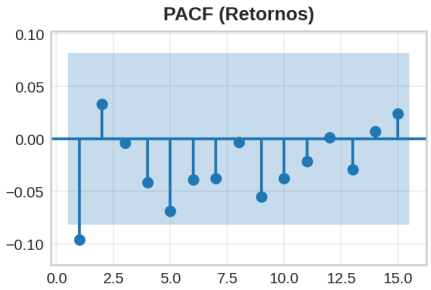
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.629
- Christoffersen (Cond.): p=0.863

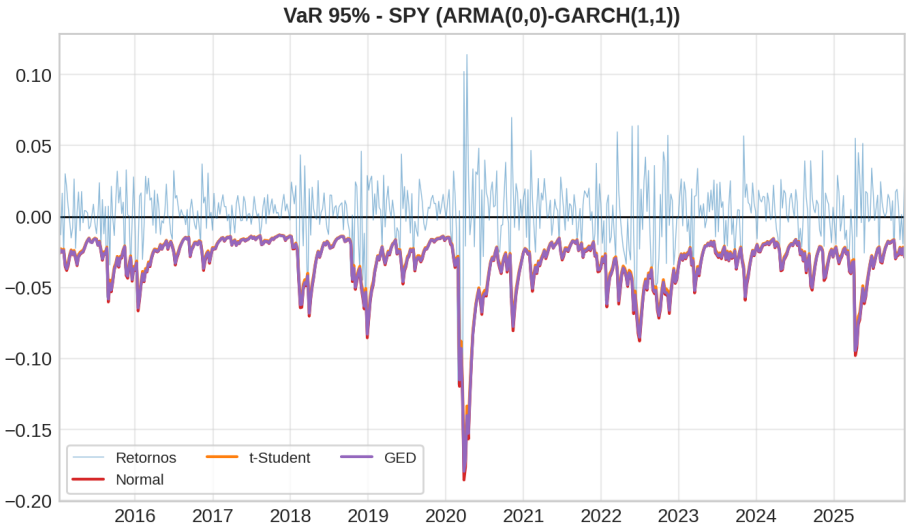
SPY

S&P; 500: Benchmark de Mercado.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 44/28
- **Tasa de Fallo:** 7.73% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	2412	0.358	0.642	0.364	nan
t-Student	2376	0.311	0.665	0.395	6.24
GED	2385	0.328	0.655	0.397	1.37

Serie: SPY
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.00
Volatilidad	0.17
Sharpe	0.53
VaR 95	-3.38%
VaR 99	-6.05%

Fundam.	
Price	$695.49
PE	28.2
Target	$None
Upgrade	-

Nota: Referencia.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.2700
- ARCH-LM (H.Cond): p=0.0000

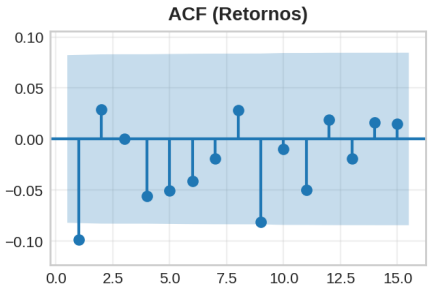
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.976 (<1)
- AIC: 2376
- Ljung-Box (Resid. Std.): p=0.522
- ARCH-LM (Resid. Std.): p=0.851

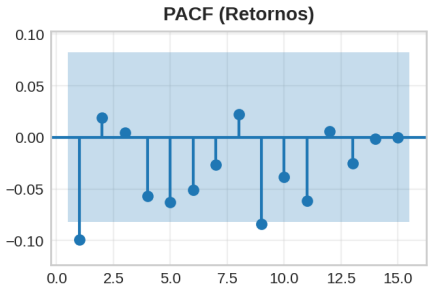
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.005
- Christoffersen (Cond.): p=0.014

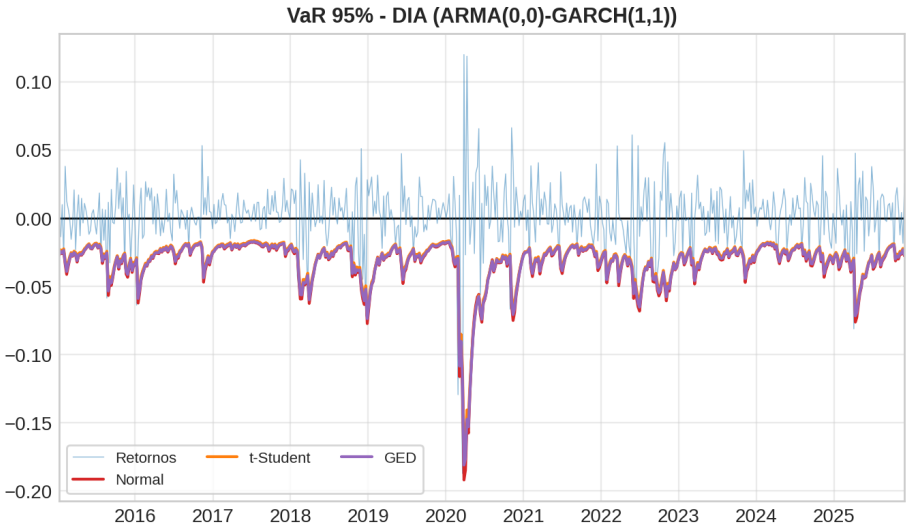
DIA

Dow Jones: Industriales Blue-Chip.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 41/28
- **Tasa de Fallo:** 7.21% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	2420	0.281	0.650	0.293	nan
t-Student	2372	0.234	0.699	0.324	5.15
GED	2378	0.243	0.682	0.328	1.26

Serie: DIA
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	0.96
Volatilidad	0.17
Sharpe	0.43
VaR 95	-3.12%
VaR 99	-6.47%

Fundam.	
Price	$490.06
PE	23.7
Target	$None
Upgrade	-

Nota: Value.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.1344
- ARCH-LM (H.Cond): p=0.0000

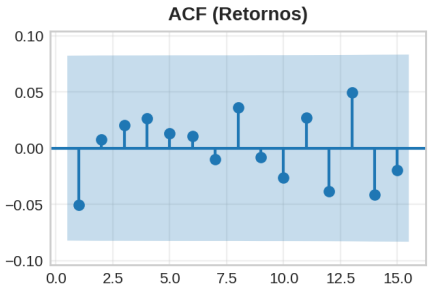
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.932 (<1)
- AIC: 2372
- Ljung-Box (Resid. Std.): p=0.473
- ARCH-LM (Resid. Std.): p=0.734

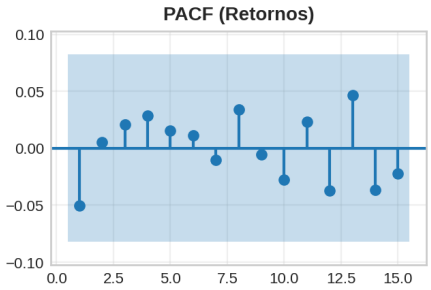
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.023
- Christoffersen (Cond.): p=0.062

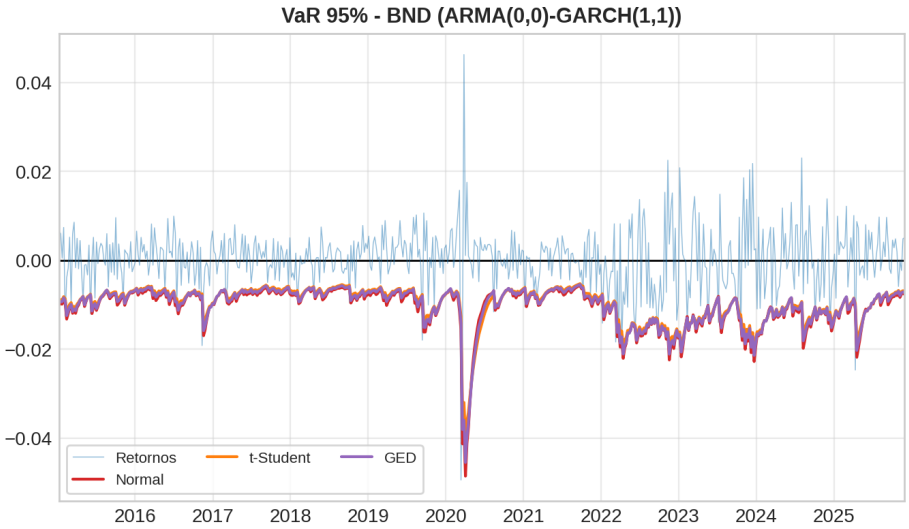
BND

Total Bond: Renta Fija.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 35/28
- **Tasa de Fallo:** 6.15% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	1071	0.230	0.718	0.027	nan
t-Student	1043	0.155	0.805	0.061	6.35
GED	1050	0.194	0.758	0.058	1.39

Serie: BND
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	0.06
Volatilidad	0.05
Sharpe	-0.35
VaR 95	-1.06%
VaR 99	-1.84%

Fundam.	
Price	$74.28
PE	0.0
Target	$None
Upgrade	-

Nota: Refugio.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.9653
- ARCH-LM (H.Cond): p=0.0000

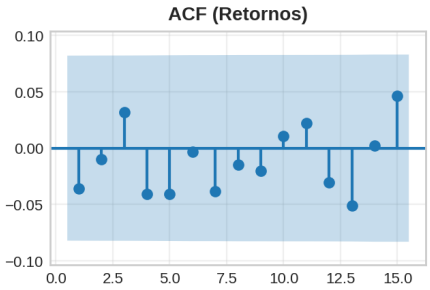
ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.960 (<1)
- AIC: 1043
- Ljung-Box (Resid. Std.): p=0.784
- ARCH-LM (Resid. Std.): p=0.442

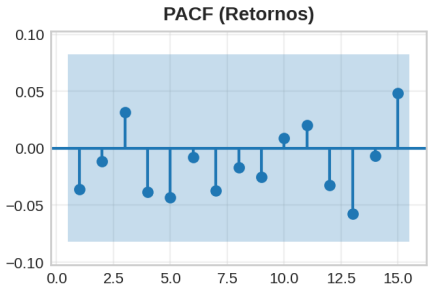
ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.223
- Christoffersen (Cond.): p=0.310

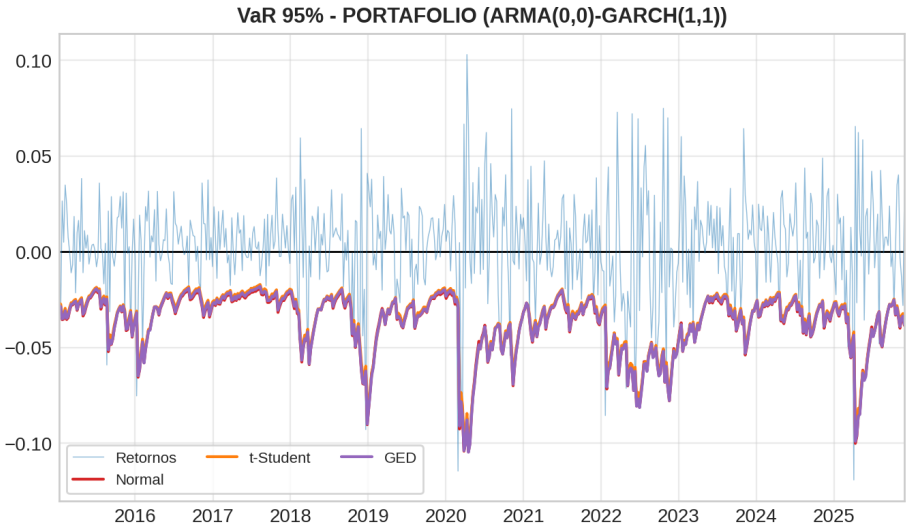
PORTAFOLIO

PORTAFOLIO: Diversificación Estratégica.

- **Modelo Óptimo:** ARMA(0,0)-GARCH(1,1) con t-Student
- **AC en Residuos:** ■ Eliminada
- **Violaciones VaR:** 41/28
- **Tasa de Fallo:** 7.21% (Obj: 5%)







Parámetros ARMA(0,0)-GARCH(1,1) Estimados

Model	AIC	Alpha	Beta	Media	Shape (nu)
Normal	2626	0.221	0.721	0.483	nan
t-Student	2592	0.196	0.762	0.502	6.31
GED	2598	0.213	0.738	0.513	1.35

Serie: PORTAFOLIO
Rango: 2015-01-01 a 2025-11-27

Métricas (Anualizadas)	
Beta	1.01
Volatilidad	0.19
Sharpe	0.70
VaR 95	-4.12%
VaR 99	-7.56%

Fundam.	
Price	-
PE	-
Target	-
Upgrade	-

Nota: Eficiente.

ETAPA 1: Diagnóstico (Retornos)
- ADF (Estacionario): p=0.0000
- Ljung-Box (AC Pre-Mod): p=0.9170
- ARCH-LM (H.Cond): p=0.0000

ETAPA 2: Ajuste ARMA(0,0)-GARCH(1,1)
- Distribución: **t-Student**
- Suma $\alpha + \beta$: 0.958 (<1)
- AIC: 2592
- Ljung-Box (Resid. Std.): p=0.979
- ARCH-LM (Resid. Std.): p=0.844

ETAPA 3: Backtesting (VaR 95%)
- Kupiec (No Cond.): p=0.023
- Christoffersen (Cond.): p=0.029